Ogbonna Arinze

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SUMMARY

Full Stack Engineer with 8 years of experience designing and delivering scalable, production-ready applications across healthcare, fintech, civic tech, and enterprise environments. Proven expertise in AI integration, including production RAG pipelines, multi-agent LLM orchestration, and vector search systems, combined with deep full-stack capabilities in React, Next.js, Node.js, NestJS, and PostgreSQL. Experienced in architecting cloud-native distributed systems with Docker, CI/CD automation, and infrastructure-as-code. Adept at leading technical teams, making architectural decisions, and coaching developers across cross-functional consulting engagements.

CORE COMPETENCIES

- **AI & LLM Systems:** RAG pipelines, multi-agent orchestration (Mastra), OpenAI API, LangChain, Vercel AI SDK, vector search (pgvector, Chroma), embeddings (Voyage AI), Cohere reranking, intent classification, and agentic workflows.
- **Full-Stack Development:** React, Next.js, TypeScript, Node.js, NestJS, REST APIs, GraphQL, JWT authentication, RBAC, SSR/SSG, and monorepo architecture (Nx).
- **Cloud & Infrastructure:** AWS (CloudFront, S3), Docker/Docker Compose, GitHub Actions, CI/CD pipelines, Traefik reverse proxy, multi-stage Dockerfiles, and health-checked deployments.
- **Databases & Data:** PostgreSQL (pgvector), MongoDB, Prisma, data visualization (Recharts, react-chartjs), real-time streaming (SSE, EventSource API), and BM25 full-text search.
- **Testing & Quality:** Playwright, Jest, Cypress, Allure Reports, Storybook, ESLint, Prettier, and CI-driven automated QA.
- **Architecture & DevOps:** Distributed systems design, API architecture, system integration, cloud-native patterns, code splitting, performance optimization, and bundle analysis.
- **Leadership:** Technical direction, architecture reviews, developer mentorship, Agile/Scrum, TDD/BDD, and cross-industry consulting adaptability.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)

Frontend Engineer (React / AI Integration) *May 2023 - Present*

- Co-developed a full-stack **Next.js chat widget** powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to connect data sources and query them in real time via a conversational interface.
- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension**, optimizing workflows for clinical site integrations with upstream EHR systems.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Implemented **JWT-based authentication** and **role-based access control** in Next.js with TypeScript and React Query, using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Optimized application performance by implementing **lazy loading, code splitting, dynamic imports**, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality.

Descasio Lagos, Nigeria (Hybrid)

Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application.
- Enhanced development workflow by introducing **Jest** and **Enzyme** automated testing frameworks, leading to a **20% decrease** in bug reports and establishing a culture of test-driven development.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes across the team.

- Utilized **Storybook** for documenting React components, streamlining change management and supporting agile development practices.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3, handling thousands to millions of daily transactions and boosting settlement rates by **80%**, serving finance industry clients.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React, Redux, RxJS**, and **Axios** for asynchronous API integration.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Adopted the native **EventSource API** for real-time data streaming, replacing socket.io for faster updates and a more responsive interface.
- Built dynamic data visualization dashboards using **React** and **react-chartjs** with **Redux Toolkit** and **Ant Design**, streamlining management of user roles and permissions.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, focusing on testable, maintainable, and well-documented code.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** (testing) and **AWS** (production).
- Mentored two interns through frontend and backend projects, providing technical guidance and professional development support.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app (22 interactive chart types with Recharts), Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service, all orchestrated with pnpm workspaces.
- Designed and implemented a **multi-agent AI system** using the **Mastra** framework with **5 specialized agents** (Router, Budget Analyst, Corruption Analyst, GovSpend Analyst, Impact Analyst) and a three-stage intent classification pipeline (keyword scoring, LLM classification with 5-min cache, inferred fallback) for domain-specific **RAG-powered analysis**.
- Built a **hybrid RAG pipeline** combining **pgvector** similarity search with **BM25** full-text search, merged via Reciprocal Rank Fusion (K=60), with **Cohere** rerank-2 reranking and **Voyage AI** voyage-3-large embeddings (1024-dim, 708K+ chunks across 4 vector indexes).
- Built the document ingestion pipeline supporting PDF (with OCR via Tesseract), XLSX/XLS, DOCX, JSON, and Markdown extractors, processing **700+ documents** (959 files) across 37 Nigerian states with zero ingestion errors.
- Configured **CI/CD** with **GitHub Actions** for automated builds and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits (API 1GB, Ingest 4GB, Postgres 2GB), and 30s graceful shutdown. Integrated **Traefik** for reverse proxy routing.
- Implemented **SSE streaming** for real-time chat responses with 15s keepalive heartbeats, **PostgreSQL 16** with **Prisma v7**, phone OTP and Telegram OAuth authentication, and shareable public conversations with SEO optimization (Open Graph, JSON-LD, SSR).
- Integrated **Langfuse** for LLM observability and scoring, **Infisical** for secrets management across environments, and **Playwright** E2E tests across 4 test projects (web auth, dashboard auth, desktop Chrome, mobile Pixel 5).

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*